AquaNode R0.3-EC-revB

External connection review / 29 October 2026

BENCH ONLY - NOT A NATIVE KICAD SCHEMATIC

This document summarizes the revised external harness. Native schematic export, ERC/DRC, firmware compilation and physical validation are pending. Do not manufacture from this document.

From	To / action
TP5 / 5V_PH	SEN0244 VCC via insulated external wire
J7.2 / 3V3	NO cable contact; never bridge to TP5
SEN0244 GND	J7.1 / common ground
SEN0244 A	R21 1k -> J7.3 -> R4 100R -> J1.1 / GPIO1
C19 100nF	J1.1 / EC_ADC to GND, close to MCU
R3	DNP / REMOVE
DS18B20 DQ	R20 100R -> J1.7 / GPIO43
R19 4.7k	DS_LOCAL / GPIO43 to 3V3
D3 ESD	DS_CABLE to GND
DS18B20 VDD / GND	3V3 / GND, externally powered
J9 / pH probe	Disconnect module; remove probe from sample
GPIO2 / PH_EN	HIGH enables existing shared 5V boost/LDO
ADS1115 AIN2 / AIN3	Remain GND; DO NOT attach EC
Wio radio	Stock B2B only; not used by bench firmware

Source of truth: design.json + cad/ + EC_integration_ru.md. Capacitor and protection parts are on the external assembly, not the carrier PCB. Retain original project attribution and licences; see ATTRIBUTION.md.